

Multimodal Emotion Recognition — CNN + SVM Fusion

A university ML project that predicts emotion (angry, happy, sad) by fusing facial expression analysis (CNN) with speech analysis (SVM).

Overview

The system combines two independently trained models:

- **CNN** — classifies emotion from a face image (48×48 px)
- **SVM** — classifies emotion from audio features (MFCCs, pitch, energy)

Their probability outputs are blended using optimised weights (**W_CNN**, **W_SVM**) to produce a final prediction.

Project Structure

File	Purpose
<code>Full_model_AI.ipynb</code>	Train the CNN on face image dataset
<code>speech_emotion_recognition.ipynb</code>	Train the SVM on voice dataset
<code>Weight finder.ipynb</code>	Optimise CNN/SVM fusion weights via grid search + scipy
<code>4_Predictor.ipynb</code>	Run predictions on file pairs; generate evaluation report
<code>5_Live.ipynb</code>	Real-time webcam + microphone emotion detection
<code>1_CNN_model.h5</code>	Saved CNN model
<code>2_SVM_model.pkl</code>	Saved SVM model
<code>3_Combined_model.pkl</code>	Saved fusion weights
<code>scaler.pkl</code>	StandardScaler fitted on audio training data

Datasets

dataset/ # Face images — subfolders: angry/ happy/ sad/

dataset_voice/ # Audio files — subfolders: angry/ happy/ sad/

Audio formats supported: `.wav`, `.mp3`, `.flac`

Image formats supported: `.jpg`, `.jpeg`, `.png`

Setup

```
pip install tensorflow scikit-learn librosa opencv-python \
```

```
    joblib sounddevice scipy Pillow matplotlib numpy
```

Usage

Training

1. Train CNN: run all cells in `Full_model_AI.ipynb`
2. Train SVM: run all cells in `speech_emotion_recognition.ipynb`
3. Find optimal fusion weights: run `Weight finder.ipynb`

Prediction (file input)

Open `4_Predictor.ipynb` and set the paths at the bottom:

```
AUDIO_FILE = "dataset_voice/Happy/1047_IWL_HAP_XX.wav"
```

```
IMAGE_FILE = "dataset/happy/265.jpg"
```

Run all cells. Output includes per-model probabilities, fused prediction, confidence, and a visualisation.

Live Detection

Run all cells in `5_Live.ipynb`. Controls:

- `SPACE` — force an immediate prediction
- `Q` — quit

Model Details

CNN

- Input: 48×48 RGB face images, normalised to [0, 1]
- 80/20 train/validation split
- Output: softmax probabilities over 3 emotion classes

SVM

- Features: 13 MFCCs (mean) + mean pitch (YIN) + mean RMS energy = 15-dim vector
- Features scaled with `StandardScaler`
- Probability estimates enabled (`predict_proba`)

Fusion

- Weighted sum: `combined = W_CNN × cnn_probs + W_SVM × svm_probs`
- Weights determined by grid search over validation pairs, refined with `scipy.optimize.minimize`

Emotions

`angry · happy · sad`